

Rajarshi Shahu College of Engineering

IEEE Research Internship

Topic: Loan Approval Prediction

Group Members:

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Week 2 – Development & Experimental Setup

Objectives

- Begin implementation of the loan approval prediction system
- Prepare and preprocess the dataset for model training
- Set up the experimental environment with required tools and libraries
- Develop initial machine learning models for baseline comparison
- Perform Exploratory Data Analysis (EDA) to understand data distributions

Activities

- Data loading, inspection, and type casting
- Exploratory Data Analysis – histograms, boxplots, pie charts, count plots
- Feature classification into numerical and categorical columns
- Class distribution analysis of target variable (loan_status)
- Outlier detection using IQR-based boxplots; trimming with OutlierTrimmer
- Feature scaling using StandardScaler (skewed) and MinMaxScaler (normal)
- Label encoding of all categorical variables
- Correlation analysis for feature selection (threshold > 0.1)
- Train-test split (80:20) and model training
- Baseline model development: Logistic Regression, SVC, KNN
- Evaluation using Accuracy, Classification Report, Confusion Matrix

Expected Deliverables

- Preprocessed Dataset
- Source Code (Initial Version – Jupyter Notebook)
- Experimental Setup Documentation
- Preliminary Results / Outputs – Model Accuracy Scores

- EDA Graphs and Analysis (Histograms, Boxplots, Heatmaps, Pie Charts)
- Mid-Internship Progress Update (Poster & Copyright with guide name and group members' names)
- Submission of Empathy Map – Template from PBL website

Empathy Map Template: <https://docs.google.com/uc?id=1Y85fLwYffjW-iMq5BJGxKf-DbRyHGI5m&export=download>

1. Data Preprocessing & Cleaning

The dataset was loaded from `loan_data.csv` containing 45,000 records and 14 attributes. Initial inspection was carried out to identify data types, missing values, and class balance. The preprocessing pipeline involved the following steps:

1.1 Dataset Overview

Parameter	Details
Dataset Name	Loan Approval Classification Dataset
Source	Kaggle (CSV Format)
Total Records	45,000
Features	13 Input Features + 1 Target Variable (<code>loan_status</code>)
Missing Values	None Found
Data Types	Mixed – Numerical and Categorical
Tools Used	Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn

1.2 Data Type Correction

The `person_age` column was originally loaded as a float and was cast to integer type to ensure correctness:

```
df["person_age"] = df["person_age"].astype("int")
```

1.3 Feature Classification

Columns were categorised into numerical and categorical types for appropriate processing:

Feature Type	Columns
Categorical	<code>person_gender</code> , <code>person_education</code> , <code>person_home_ownership</code> , <code>loan_intent</code> , <code>previous_loan_defaults_on_file</code>
Numerical	<code>person_age</code> , <code>person_income</code> , <code>person_emp_exp</code> , <code>loan_amnt</code> , <code>loan_int_rate</code> , <code>loan_percent_income</code> , <code>cb_person_cred_hist_length</code> , <code>credit_score</code>

1.4 Missing Value Analysis

Null value check was performed using `df.isnull().sum()`. No missing values were found across all 14 columns, confirming the dataset is clean and ready for further processing.

1.5 Feature Scaling

Two scaling strategies were applied based on the distribution characteristics of each feature:

- StandardScaler applied to skewed numerical columns: `person_age`, `person_income`, `person_emp_exp`, `loan_amnt`, `loan_percent_income`, `cb_person_cred_hist_length`, `credit_score`
- MinMaxScaler applied to normally distributed column: `loan_int_rate`

1.6 Categorical Encoding

All categorical features were label-encoded using manual mapping dictionaries to preserve meaningful ordinal relationships:

Feature	Encoding Mapping
person_gender	male → 0, female → 1
person_home_ownership	RENT → 0, OWN → 1, MORTGAGE → 2, OTHER → 3
loan_intent	PERSONAL → 0, EDUCATION → 1, MEDICAL → 2, VENTURE → 3, HOMEIMPROVEMENT → 4, DEBTCONSOLIDATION → 5
previous_loan_defaults_on_file	No → 0, Yes → 1
person_education	High School → 0, Associate → 1, Bachelor → 2, Master → 3, Doctorate → 4

1.7 Outlier Detection and Removal

Boxplots were generated for all numerical features before and after outlier removal. The `OutlierTrimmer` from the `feature-engine` library was used with the IQR method (right tail) to remove extreme outliers and improve data quality.

2. Exploratory Data Analysis (EDA)

Comprehensive EDA was performed to understand the distribution of features, relationships between variables, and the class balance of the target variable.

2.1 Target Variable Distribution

A pie chart was generated to visualise the proportion of approved vs. rejected loan applications. The dataset shows a moderate class imbalance which was noted for potential SMOTE application in subsequent weeks.

Class	Label	Proportion (Approx.)
0	Loan Rejected	~78%
1	Loan Approved	~22%

2.2 Numerical Feature Distribution

Histograms were plotted for all numerical columns using a 20-bin configuration (figure size 12x7). Observations:

- person_income and loan_amnt exhibit right-skewed distributions
- credit_score follows an approximately normal distribution
- person_emp_exp and cb_person_cred_hist_length show exponential-like distributions

2.3 Categorical Feature Analysis

Count plots were generated for each categorical variable using a custom plot_categorical_column function. A 5% threshold line was added to highlight minority categories. Key observations:

- person_gender: dataset is roughly balanced between male and female applicants
- loan_intent: EDUCATION and MEDICAL are the most frequent loan purposes
- person_home_ownership: RENT and MORTGAGE are predominant
- previous_loan_defaults_on_file: majority of applicants have no prior defaults

2.4 Correlation Heatmap & Feature Selection

A correlation matrix was computed on the cleaned dataset (df2) and visualised using a Seaborn heatmap with the 'coolwarm' colour map. Features with absolute correlation with loan_status greater than 0.1 were selected for model training:

Selected Feature	Correlation with loan_status
credit_score	High negative (better score → more approvals)
previous_loan_defaults_on_file	High positive (defaults → rejections)
loan_percent_income	Moderate positive
person_income	Moderate negative
loan_int_rate	Moderate positive
loan_amnt	Low positive

person_emp_exp	Low negative
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3. Model Development & Training

Three baseline machine learning models were developed and evaluated on the selected feature set to establish initial performance benchmarks. The dataset was split into 80% training and 20% testing sets.

```
Train-Test Split: X_train, X_test, y_train, y_test = train_test_split(X_selected, Y,
test_size=0.2, random_state=42)
```

3.1 Logistic Regression

A Logistic Regression model was trained with max_iter=1000 and random_state=42 to ensure convergence and reproducibility.

Metric	Training Set	Test Set
Accuracy	Computed via model.score()	Computed via accuracy_score()
Model Parameters	max_iter=1000, random_state=42	–
Status	Trained & Evaluated	–

3.2 Support Vector Classifier (SVC)

An SVC model with default parameters was trained and evaluated. A confusion matrix heatmap was generated to visualise true/false positive and negative rates.

3.3 K-Nearest Neighbours (KNN)

KNN was trained with k=3. A detailed Classification Report (Precision, Recall, F1-Score per class) was generated. The confusion matrix heatmap was plotted for visual evaluation. The trained KNN model was also saved using Pickle for future use.

```
Model Saved: knn_model.pkl (saved using pickle)
```

3.4 Model Comparison Summary

Model	Key Parameters	Evaluation Performed	Status
Logistic Regression	max_iter=1000	Accuracy Score	Completed

SVC	Default	Accuracy + Confusion Matrix	Completed
KNN	k=3	Accuracy + Classification Report + Confusion Matrix	Completed
XGBoost	TBD (Week 3)	Pending	Planned

4. Observations & Key Findings

- Dataset is clean with no missing values and does not require imputation.
- Moderate class imbalance observed (~78% rejected, ~22% approved) — SMOTE will be applied in Week 3 if required.
- credit_score and previous_loan_defaults_on_file are the strongest predictors of loan approval based on correlation analysis.
- Outlier trimming using IQR significantly reduced the influence of extreme values in person_income and loan_amnt.
- All three baseline models were trained and evaluated successfully, establishing a performance benchmark.
- KNN (k=3) model saved as knn_model.pkl for comparative evaluation in upcoming weeks.
- Correlation-based feature selection reduced dimensionality while retaining the most informative features.

5. Weekly Execution Plan – Updated Progress

Week	Planned Activities	Expected Outcome	Status
Week 1	Literature survey, research gap analysis, dataset study, methodology design, system architecture	Completed literature survey, research gaps, dataset study, methodology and system architecture	Completed
Week 2	Data preprocessing, EDA, feature selection, baseline model training (LR, SVC, KNN), evaluation	Clean dataset, initial models, accuracy scores, EDA graphs	Completed
Week 3	SHAP explainability, feature importance, risk score module, recommendation engine, fairness analysis, dashboard prototype	Explainable loan prediction system with SHAP visualisations and risk scoring	Planned
Week 4	Hyperparameter tuning, final evaluation, dashboard, PDF report, workflow automation, documentation	Final deployable loan approval prediction system	Planned

6. Additional Progress Tracking

Date	Work Completed	Challenges Faced	Next Plan
Week 2	Data preprocessing, EDA, categorical encoding, outlier removal, feature scaling, correlation analysis, baseline model training (LR, SVC, KNN), confusion matrices, classification reports	Class imbalance noted (~78:22 split); SVC training time relatively higher on full feature set	Apply SMOTE if required, train XGBoost, integrate SHAP explainability in Week 3

Mentor Approval

Dr. Seema Kedar

Signature: _____ Date: _____